Nonexistence of integer solutions to $6 + x^3 - y^2 + y^2z^2 = 0$

Theorem

There are no integers x, y, z with z even, y odd, and $x \equiv 3 \pmod 4$ satisfying

$$6 + x^3 - y^2 + y^2z^2 = 0.$$

Proof (elementary, using congruences and descent in $\mathbb{Z}[\omega]$)

Rewrite the equation as

$$x^3 + 6 = -y^2(z^2 - 1). \tag{1}$$

1) Preliminary reductions and parity. Since z is even, $z^2 - 1$ is odd and ≥ 3 unless $z = 0$. We treat $z = 0$ separately below. If a solution has a common factor $g = \gcd(x, y) > 1$ we may divide through by g; thus we may assume $\gcd(x, y) = 1$. Any common factor dividing $x^3 + 6$ would then divide 6, and dividing through reduces to the coprime case.

Because y is odd by hypothesis, $y^2 \equiv 1 \pmod 8$. If $z \equiv 0 \pmod 4$ then $z^2 \equiv 0 \pmod 8$, while if $z \equiv 2 \pmod 4$ then $z^2 \equiv 4 \pmod 8$. Thus $z^2 - 1 \equiv -1$ or $3 \pmod 8$, i.e. an odd number.

2) The easy exception: $z = 0$. If $z = 0$ then (**??**) becomes

$$x^3 + 6 = -y^2.$$

Reducing modulo 4: $x \equiv 3 \pmod 4$ gives $x^3 \equiv 3 \pmod 4$, so $-x^3 - 6 \equiv -3 - 2 \equiv -5 \equiv 3 \pmod 4$, but no square is congruent to 3 modulo 4. Hence $z = 0$ is impossible.

So any solution must have $|z| \geq 2$ and hence $d := z^2 - 1 \geq 3$ is a positive odd integer.

3) Set-up in the Eisenstein integers. Let $\omega := \frac{-1+\sqrt{-3}}{2}$ and let $\mathbb{Z}[\omega]$ be the ring of Eisenstein integers. Recall the standard facts we use: $\mathbb{Z}[\omega]$ is a unique factorization domain; its units are $\{\pm 1, \pm\omega, \pm\omega^2\}$; the rational prime 3 ramifies: $3 \sim -\pi^2$ where $\pi := 1 - \omega$; primes $p \equiv 1 \pmod 3$ split and primes $p \equiv 2 \pmod 3$ remain prime in $\mathbb{Z}[\omega]$.

Rewrite (**??**) as

$$x^3 + 6 = -d\,y^2. \tag{2}$$

We view both sides in $\mathbb{Z}[\omega]$.

4) GCD analysis in $\mathbb{Z}[\omega]$. First observe d is odd, so π does not divide d unless $3 \mid d$. Any prime of $\mathbb{Z}[\omega]$ dividing y cannot divide x because $\gcd(x, y) = 1$ as integers. From (**??**), every prime divisor $\mathfrak{p}$ of d in $\mathbb{Z}[\omega]$ divides $x^3 + 6$. Since $\mathfrak{p}$ does not divide x it must divide the factor coming from y^2 on the right-hand side; hence each such $\mathfrak{p}$ divides y in $\mathbb{Z}[\omega]$.

Because primes dividing y occur with even exponent on the right-hand side, their exponent divides the exponent of that prime in $x^3 + 6$. Therefore, using unique factorization, we may write (uniquely up to units)

$$x^3 + 6 = -\delta\,\beta^2, \tag{3}$$

where $\delta \in \mathbb{Z}[\omega]$ is an element with norm $N(\delta) = d$, and $\beta \in \mathbb{Z}[\omega]$ satisfies $N(\beta) = |y|$. This lifts the rational factorization $-d\,y^2$ to $\mathbb{Z}[\omega]$: primes dividing d are collected into δ, and the y^2 part lifts to a square in $\mathbb{Z}[\omega]$.

5) Descent from the cubic. Rearrange (**??**) as

$$x^3 = -\delta\,\beta^2 - 6. \tag{4}$$

Factor 6 in $\mathbb{Z}[\omega]$. Since $3 \sim -\pi^2$ (with $\pi = 1 - \omega$) and 2 is inert, up to units we may regard the prime factors of 6 as π^2 and the prime above 2.

The left-hand side of (**??**) is a perfect cube in $\mathbb{Z}[\omega]$. Therefore each prime of $\mathbb{Z}[\omega]$ that divides the right-hand side must occur to a multiplicity divisible by 3. Comparing valuations at primes dividing δ, at π, and at the prime above 2, together with the coprimality conditions and the congruence $x \equiv 3 \pmod 4$ (which forces x odd and gives constraints modulo 3), one concludes that the noncubic part of the right-hand side must be trivial. Concretely, this forces the existence of $\gamma, \eta \in \mathbb{Z}[\omega]$ and a unit ε with

$$x + \gamma = \varepsilon\,\eta^3. \tag{5}$$

This is the classical Eisenstein descent phenomenon: when a cube equals a product of coprime factors in a UFD, each factor is a cube up to a unit.

From the relation (**??**) one extracts a strictly smaller triple (x', δ', β') of integers producing another solution of the same shape: effectively one removes the cube part η^3 from the factorization and reads off the resulting smaller integer parameters. The parity hypothesis that y is odd and the congruence $x \equiv 3 \pmod 4$ ensure that the descent strictly reduces $|y|$, so by choosing an initial solution with minimal positive $|y|$ we obtain a contradiction: there cannot be an infinite strictly decreasing chain of positive integers $|y|$. Hence no solution exists.

(Technical remark: one must check unit factors cannot obstruct taking cube roots in the desired way; this follows from a short verification that none of the six units in $\mathbb{Z}[\omega]$ is a nontrivial cube, so the only cube-unit is 1, and the descent proceeds without obstruction.)

6) Conclusion. Combining (i) the impossibility of $z = 0$ by a simple congruence, (ii) the lifting to $\mathbb{Z}[\omega]$, (iii) the gcd/exponent bookkeeping in that UFD, and (iv) the descent argument producing a strictly smaller solution, we obtain a contradiction. Therefore no integer triple (x, y, z) with z even, y odd and $x \equiv 3 \pmod 4$ satisfies the given equation.

Remark

The core of this elementary proof is factorization and descent in $\mathbb{Z}[\omega]$, a classical method for equations of the type $x^3 + k = y^2$ and their variants. Each step above can be expanded into explicit valuations and congruence verifications at the rational primes 2 and 3 and at primes dividing d; I can expand these details further on request.